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Proposal for Resource Optimization in the Annual Operational Planning of the Faculty of Chemical Sciences of the University of Cuenca

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Abstract

This degree project develops a strategy to optimize the allocation of resources of the AOP of the Faculty of Chemical Sciences. A goal programming model based on historical data was constructed, integrating criteria of academic impact, strategic alignment, budgetary efficiency, and risk. The results show that technical prioritization improves operational efficiency and the rational use of public expenditure without increasing the available budget.

Objective

General Objective

To propose a strategy to optimize resource allocation within the POA, considering the requirements of laboratory supplies for the different university programs of the Faculty of Chemical Sciences, focused on improving efficiency and operational planning.

Specific Objective

To identify and analyze essential factors such as the allocation of laboratory supplies for operational feasibility, allowing improvement in resource allocation.

Specific Objective

To carry out data collection of the POA, analyzing budget execution and resource allocation over the last three years in the Faculty of Chemical Sciences.

Specific Objective

To develop and apply an optimization model appropriate to the POA and its simulation, allowing resource allocation to be improved efficiently and aligned with institutional objectives, and to compare its performance against the current plan.

Methodology

1. Diagnosis and analysis of the current situation, bibliographic review in databases such as Scopus and Dialnet, and review of articles in scientific journals.
2. Selection of variables, optimization method, and model development, through the diagnosis of input requirements within the POA.
3. Validation of the model by evaluating it through an analysis of allocation efficiency and resource savings, comparing its performance.

Curricular Integration

- Programming languages (programming in Lingo software)
- General economics (POA, budget allocation)
- Operations research (handling of the mathematical model, goals, and constraints)

Applied Engineering

- ISO 9001:2015 (Quality Management System)
- ISO 31000:2018 (Risk Management)
- ISO 55000:2014 (Asset Management)
- COSO ERM (2017) Internal Control and Risk Management

Keywords

Annual Operating Plan · Resource optimization · Programming by goals · Higher education.

Results

Historical data from the approved 2023 POA of the Faculty of Chemical Sciences were used, including budget distribution by projects/laboratories, Program 82 – Academic Training and Management, OEI-4, and budget execution tables, serving as a real basis for the first simulation.

The proposed model uses as decision variables the amounts allocated to laboratories, supplies, internal projects, and operational budget items linked to Program 82: Academic Training and Management. Based on these variables, goals related to the satisfaction of requirements and alignment with the Institutional Strategic Objective OEI-4 were defined.

The model obtained a feasible optimal solution for resource allocation. It also shows budget increases for the Biochemistry laboratories of 20% and Chemical Engineering of 13.5%, associated with their greater academic impact, strategic alignment, and lower risk, while Environmental Engineering and Industrial Engineering present reductions of 20.0% due to a lower relative priority in academic impact and institutional alignment.

Conclusion

Conclusion 1

The development of a strategy based on goal programming made it possible to optimize the allocation of resources of the POA of the Faculty of Chemical Sciences of the University of Cuenca, integrating academic, strategic, operational, and financial criteria. This approach proved to be suitable for supporting budgetary decision-making in a context of financial constraints typical of a public higher education institution.

Conclusion 2

An efficient allocation of resources does not depend exclusively on the available budget amount, but on the identification and prioritization of critical factors that ensure the operational viability of the POA. Among these factors are the minimum operability of laboratories, the availability of validated and traceable information, and alignment with institutional strategic objectives, particularly OEI-4.

Conclusion 3

The analysis made it possible to determine that budgetary efficiency should be evaluated beyond the execution percentage, since high execution does not guarantee efficient use of resources if the expected academic and institutional results are not achieved. In this regard, the incorporation of academic impact indicators and real execution capacity allowed for a more comprehensive evaluation of POA performance.

Conclusion 4

The inclusion of budgetary risk and internal control mechanisms within the optimization model contributed to improving the financial sustainability of the POA and reducing potential deviations in its execution. The proposed strategy demonstrates that it is possible to improve efficiency and the rational use of public expenditure without increasing the available budget, by prioritizing areas with greater academic and institutional impact.

Contacts

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